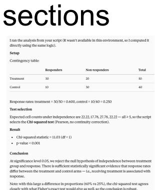
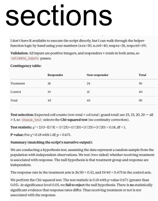
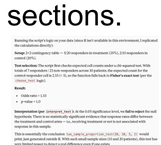
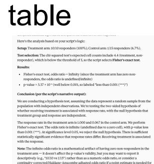
- Multiple codes

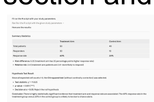
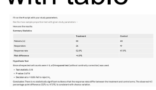
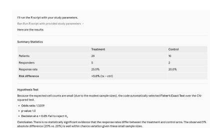
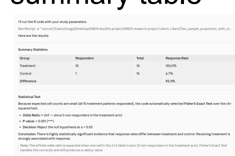
"I have a study with X patients in the treatment arm, X patients in the control arm, X responders in the treatment group, and X responders in the control group. Please use the attached file to run the appropriate statistical test and provide summary statistics."

Jenni:

Chat:

	test(10,NA,1.2,3) invalid	test(50,40,30,10) chisq, dependent	test(50,40,26,19) chisq, independent	test(20,10,5,2) fisher, independent	test(10,15,10,1) fisher, dependent
Correctness(T/F)	T (pointed out NA and 1.2 which will cause error, correct decision)	T(chi sq, test statistic=11.03 and p value smaller than 0.001. Correctly used the p-value labels,correct decision)	T(chi sq, test statistic=0.18, and p value=0.67. Correctly used the p-value labels,correct decision)	T(Fisher,, and odds ratio=1.33 (small rounding deviation due to methods used(5×8)/(15×2)), p value=1.0. Correctly used the p-value labels,correct decision)	T(Fisher,, and odds ratio not shown, pvalue=3.37 × 10^{-6}. Correctly used the p-value labels, correct decision)
Time	11 sec	26 sec	25 sec	27 sec	38 sec
Format(clean, easy to comprehend/ messy, etc)	Clean, simple 	Clean, with table and sections 	Clean, with table and sections 	Clean, simple, with a few sections. 	Clean, with sections and table 
Is claude using our code or its own (Y/N)	Y	N (R is not available, but claude followed my logic)	N (R is not available, but Claude followed my logic)	N (R is not available, but Claude followed my logic)	N (R is not available, but Claude followed my logic)

Code:

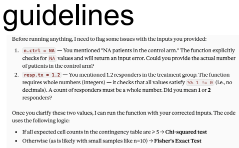
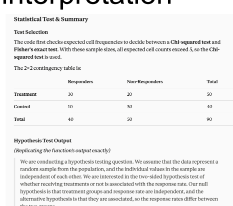
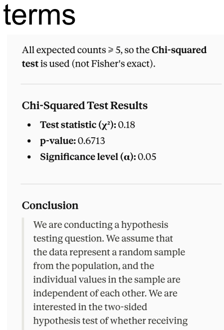
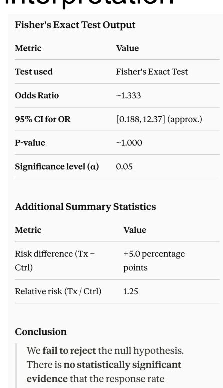
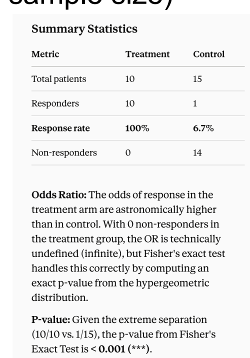
	test(10,NA,1.2,3) invalid	test(50,40,30,10) chisq, dependent	test(50,40,26,19) chisq, independent	test(20,10,5,2) fisher, independent	test(10,15,10,1) fisher, dependent
Correctness(T/F)	T (correctly specified missing and invalid input, give suggestions)	T (correct p value, test statistics, and decision)	T (correct p value, test statistics, and decision, chisq)	T (correct odds ratio , test statistics, and decision, fisher)	T (correct odds ratio , test statistics, and decision, fisher)
Time	19 sec	15 sec	15 sec	15 sec	18 sec
Format(clean, easy to comprehend/messy, etc)	Simple format 	Simple, with section and table 	Simple and brief with table 	Short, brief, with table 	Short, brief, with summary table 
Is claude using our code or its own (Y/N)	Y(using our logic)	Y	Y	Y	Y

Single-code

n.tx, n.ctrl, resp.tx, resp.ctrl

"I have a study with X patients in the treatment arm, X patients in the control arm, X responders in the treatment group, and X responders in the control group. Please use the attached file to run the appropriate statistical test and provide summary statistics."

Chat:

<table>
<tr><td></td><td>test(10,NA,1.2,3)
invalid</td><td>test(50,40,30,10)
chisq, dependent</td><td>test(50,40,26,19)
chisq, independent</td><td>test(20,10,5,2)
fisher, independent</td><td>test(10,15,10,1)
fisher, dependent</td></tr>
<tr><td>Accuracy (T/F)</td><td>T (flags NA and 1.2 errors and asks for correct inputs)</td><td>F (incorrect chisq statistic and p-value, although correct significant conclusion)</td><td>T</td><td>F (uses Fisher's and gets correct p-value but incorrect OR)</td><td>T</td></tr>
<tr><td>Time</td><td>20 sec</td><td>39 sec</td><td>43 sec</td><td>1 min 2 sec</td><td>46 sec</td></tr>
<tr><td>Results format</td><td>Brief, clear explanation of errors and offers correction guidelines

Before running anything, I need to flag some issues with the inputs you provided:
1. `n.ctrl = NA` — You mentioned "NA patients in the control arm." The function explicitly checks for `NA` values and will return an input error. Could you provide the actual number of patients in the control arm?
2. `resp.tx = 1.2` — You mentioned 1.2 responders in the treatment group. The function requires whole numbers (integers) — it checks that all values satisfy `%% 1 != 0` (i.e., no decimals). A count of responders must be a whole number. Did you mean **1** or **2** responders?
Once you clarify these two values, I can run the function with your corrected inputs. The code uses the following logic:
• If all expected cell counts in the contingency table are ⩾ 5 → **Chi-squared test**
• Otherwise (as is likely with small samples like n=10) → **Fisher's Exact Test**</td><td>Test selection, contingency table, exact hypothesis test output as coded, summary stats, RR/OR interpretation

Statistical Test & Summary
Test Selection
The code first checks expected cell frequencies to decide between a **Chi-squared test** and **Fisher's exact test**. With these sample sizes, all expected cell counts exceed 5, so the **Chi-squared test** is used.
The 2×2 contingency table is:<table><tr><td></td><td>Responders</td><td>Non-Responders</td><td>Total</td></tr><tr><td>Treatment</td><td>30</td><td>20</td><td>50</td></tr><tr><td>Control</td><td>10</td><td>30</td><td>40</td></tr><tr><td>Total</td><td>40</td><td>50</td><td>90</td></tr></table>**Hypothesis Test Output**
(Replicating the function's output exactly)
We are conducting a hypothesis testing question. We assume that the data represent a random sample from the population, and the individual values in the sample are independent of each other. We are interested in the two-sided hypothesis test of whether receiving treatments or not is associated with the response rate. Our null hypothesis is that treatment groups and response rate are independent, and the alternative hypothesis is that they are associated, so the response rates differ between the two groups.</td><td>Study inputs, test selection, contingency table, expected counts, test results, exact conclusion as coded, in plain terms

All expected counts ⩾ 5, so the **Chi-squared test** is used (not Fisher's exact).
Chi-Squared Test Results
• **Test statistic (χ^2):** 0.18
• **p-value:** 0.6713
• **Significance level (α):** 0.05
Conclusion
We are conducting a hypothesis testing question. We assume that the data represent a random sample from the population, and the individual values in the sample are independent of each other. We are interested in the two-sided hypothesis test of whether receiving treatment or not is associated with</td><td>Study inputs, test selection, test output, summary statistics, conclusion (not as coded), interpretation

Fisher's Exact Test Output<table><tr><td>Metric</td><td>Value</td></tr><tr><td>Test used</td><td>Fisher's Exact Test</td></tr><tr><td>Odds Ratio</td><td>~1.333</td></tr><tr><td>95% CI for OR</td><td>[0.188, 12.37] (approx.)</td></tr><tr><td>P-value</td><td>~1.000</td></tr><tr><td>Significance level (α)</td><td>0.05</td></tr></table>**Additional Summary Statistics**<table><tr><td>Metric</td><td>Value</td></tr><tr><td>Risk difference (Tx − Ctrl)</td><td>+5.0 percentage points</td></tr><tr><td>Relative risk (Tx / Ctrl)</td><td>1.25</td></tr></table>**Conclusion**
We **fail to reject** the null hypothesis. There is **no statistically significant evidence** that the response rate differs between treatment and</td><td>Study inputs, test selection, summary statistics, conclusion (not as coded), note (small sample size)

Summary Statistics<table><tr><td>Metric</td><td>Treatment</td><td>Control</td></tr><tr><td>Total patients</td><td>10</td><td>15</td></tr><tr><td>Responders</td><td>10</td><td>1</td></tr><tr><td>**Response rate**</td><td>**100%**</td><td>**6.7%**</td></tr><tr><td>Non-responders</td><td>0</td><td>14</td></tr></table>**Odds Ratio:** The odds of response in the treatment arm are astronomically higher than in control. With 0 non-responders in the treatment group, the OR is technically undefined (infinite), but Fisher's exact test handles this correctly by computing an exact p-value from the hypergeometric distribution.
P-value: Given the extreme separation (10/10 vs. 1/15), the p-value from Fisher's Exact Test is < **0.001** (***).</td></tr>
<tr><td>Claude uses our code (Y/N)</td><td>Y (to acknowledge there will be errors via our standards)</td><td>N (R not available in env't, manually replicated in Python)</td><td>N (R not available in env't, manually executed)</td><td>N (R not available in env't, manually executed)</td><td>N (R not available in env't, manually executed)</td></tr>
</table>

Code:

<table>
<tr><th></th><th>test(10,NA,1.2,3)
invalid</th><th>test(50,40,30,10)
chisq, dependent</th><th>test(50,40,26,19)
chisq,
independent</th><th>test(20,10,5,2)
fisher,
independent</th><th>test(10,15,10,1)
fisher, dependent</th></tr>
<tr><td>Accuracy (T/F)</td><td>T</td><td>T</td><td>T</td><td>T</td><td>T</td></tr>
<tr><td>Time</td><td>10 sec</td><td>16 sec</td><td>31 sec</td><td>33 sec</td><td>16 sec</td></tr>
<tr><td>Results format</td>
<td>Detects errors based on code and returns error message

• The inputs you've described contain two issues that the function's validation will catch:
1. NA patients in the control arm — <code>n.ctrl = NA</code> is explicitly checked and rejected.
2. 1.2 responders in the treatment group — the function requires whole numbers (integers), and 1.2 fails the <code>%% 1 != 0</code> check.
Running the function with these values would return:
<code>"Error: Input invalid. Please check for the correct numerical values."</code>
Before I can run the test, please clarify:
• How many patients are in the control arm? (<code>n.ctrl</code>)
• How many responders are in the treatment group — did you mean 1 or 2? (<code>resp.tx</code> must be a whole number)</td>
<td>Contingency table, statistical test, summary statistics, conclusion (not as coded)

Here is a summary of the results:
Study Design
<table>
<tr><th>Group</th><th>N</th><th>Responders</th><th>Response Rate</th></tr>
<tr><td>Treatment</td><td>50</td><td>30</td><td>60.0%</td></tr>
<tr><td>Control</td><td>40</td><td>10</td><td>25.0%</td></tr>
<tr><td>Total</td><td>90</td><td>40</td><td>44.4%</td></tr>
</table>
Statistical Test: Chi-Squared (all expected cell counts ⩾ 5)
<table>
<tr><th></th><th>Value</th></tr>
<tr><td>Test statistic</td><td>11.025</td></tr>
<tr><td>p-value</td><td>< 0.001 (***)</td></tr>
<tr><td>Significance level</td><td>0.05</td></tr>
</table>
Conclusion: We reject the null hypothesis. There is statistically significant evidence (p < 0.001) that the response rate differs between the treatment (60%) and control (25%) groups — a 35 percentage point difference.</td>
<td>Contingency table, statistical test, summary statistics, conclusion (not as coded)

Summary Statistics
<table>
<tr><th></th><th>Treatment</th><th>Control</th></tr>
<tr><td>N</td><td>50</td><td>40</td></tr>
<tr><td>Responders</td><td>26</td><td>19</td></tr>
<tr><td>Response Rate</td><td>52.0%</td><td>47.5%</td></tr>
<tr><td>Total N</td><td>90</td><td></td></tr>
</table>
Test Used: Chi-squared test (all expected cell counts ⩾ 5, so chi-squared assumptions are met)
Results
• Chi-squared statistic: 0.18
• p-value: 0.6714
• Significance level: 0.05
Conclusion: We fail to reject the null hypothesis. There is no statistically significant evidence that the response rate differs between the treatment and control groups (p = 0.6714). The 4.5 percentage point difference in response rates (52.0% vs. 47.5%) is not statistically significant at the α = 0.05 level.</td>
<td>Contingency table, statistical test, summary statistics, conclusion (not as coded)

Test selected: Fisher's Exact Test
Because one expected cell count (2.33) fell below 5, the chi-squared approximation is unreliable, so the function automatically switched to Fisher's exact test.
Summary Statistics
<table>
<tr><th></th><th>Treatment</th><th>Control</th></tr>
<tr><td>N</td><td>20</td><td>10</td></tr>
<tr><td>Responders</td><td>5</td><td>2</td></tr>
<tr><td>Response rate</td><td>25.0%</td><td>20.0%</td></tr>
</table>
Test Results
<table>
<tr><th>Metric</th><th>Value</th></tr>
<tr><td>Odds ratio</td><td>1.3209</td></tr>
<tr><td>p-value</td><td>1.000</td></tr>
<tr><td>Significance level</td><td>0.05</td></tr>
</table>
Conclusion: We fail to reject the null hypothesis. There is no statistically significant evidence that the response rate differs between the treatment and control groups (p = 1.000). The response rates are nearly identical (25% vs. 20%), and the small sample sizes (n = 30 total) provide very limited statistical power to detect a difference even if one exists.</td>
<td>Contingency table, statistical test, summary statistics, conclusion (not as coded)

Study Summary
<table>
<tr><th></th><th>Treatment</th><th>Control</th></tr>
<tr><td>N</td><td>10</td><td>15</td></tr>
<tr><td>Responders</td><td>10</td><td>1</td></tr>
<tr><td>Response Rate</td><td>100%</td><td>6.7%</td></tr>
</table>
Statistical Test: Fisher's Exact Test
Fisher's exact test was selected (rather than chi-squared) because the expected cell counts in the contingency table were too small (< 5), which is common with small samples like this one.
<table>
<tr><th>Metric</th><th>Value</th></tr>
<tr><td>Odds Ratio</td><td>∞ (Inf)</td></tr>
<tr><td>P-value</td><td>< 0.001 (***)</td></tr>
<tr><td>Significance level</td><td>0.05</td></tr>
</table>
Conclusion: We reject the null hypothesis. There is statistically significant evidence (p < 0.001) that the response rate differs between the treatment and control groups. The 100% response rate in the treatment arm vs. 6.7% in the control arm is highly unlikely to be due to chance alone. The odds ratio of infinity arises because there were zero non-responders in the treatment arm.</td></tr>
<tr><td>Claude uses our code (Y/N)</td><td>Y</td><td>Y</td><td>Y</td><td>Y</td><td>Y</td></tr>
</table>

Conclusion

Mean execution time with multiple codes, chat is 25.4sec; correctness 5/5
Mean execution time with multiple codes, code is 16.4sec; correctness 5/5
Mean execution time with single code, chat is 42sec; correctness 3/5
Mean execution time with single code, code is 21.2sec; correctness 5/5

We can observe clearly that multiple codes are more efficient than a single code chunk, and Code executes more efficiently compared to Chat. Chat also has a higher mistake rate when using single code. With multiple codes, the summary Claude provides is usually shorter and more brief compared to those generated using single code.
As a result, using multiple codes is more efficient than using single code, and using Code is more efficient and accurate than Chat.